

NSA Grow Your Own Food Series: Beginner's Guide to Vermicomposting

By Edible Garden City

Cardboard

safe
x died less
(may have
predator
eating &
theater)

1/keagray
bin

Worms from outside?

Topics covered

- Understand how vermicomposting helps with food waste
- Differentiate between different earthworms
- Create a basic worm bin
- List how to care for a worm bin & harvest vermicast

overcrowding?
• upgrade
• Godm.

Introduction

Bin care

Bedding

- Shredded cardboard, paper, or newspaper
- Moisture

Rotate (beds
spots)



Bedding for a worm bin consists of any light organic material that's high in carbon, which helps to keep moisture in the bin and protects the worms from drying out. Dry shredded paper adds air, which is a vital component to a healthy worm bin system; it also serves as an additional food for the worms.

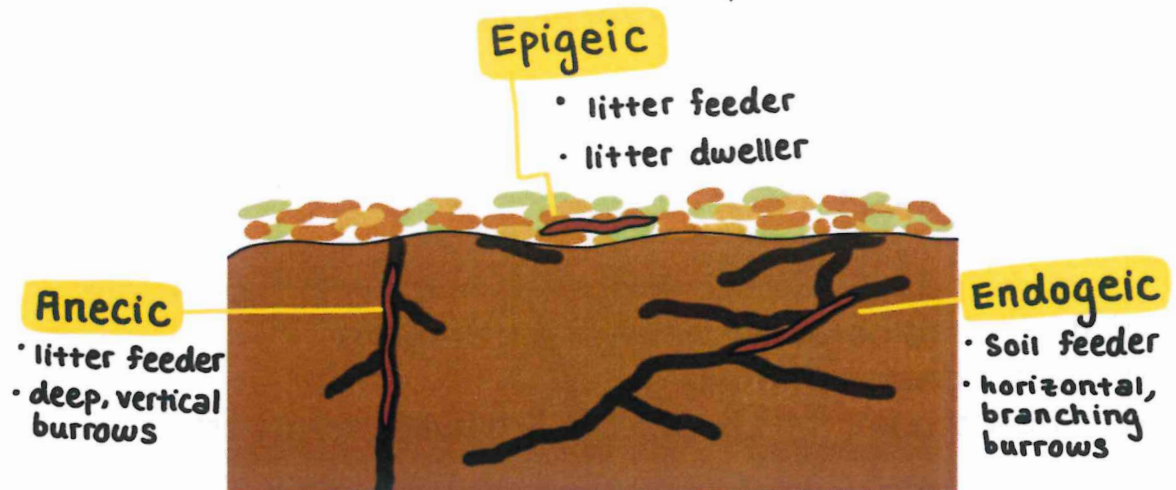
Feed 1x wk
1/4 cup

Finish up
green, tw
feed

What cannot be use:

- Plastics
- Materials that are oily or waxy
- Materials with color prints (Singapore uses soy-based ink for newspapers hence we can use them!)

Types of Earthworms:



Anecic:

Worms make permanent vertical burrows in soil. They feed on leaves on the soil surface that they drag into their burrows. They also cast on the surface, and these casts can quite often be seen in grasslands. Some anecic earthworm species also make middens (piles of casts) around the entrance to their burrows. Anecic species are the largest species of earthworms in the UK. They are darkly coloured at the head end (red or brown) and have paler tails. Burrows can be 2cm wide.

Endogeic:

Earthworms live in and feed on the soil. They make horizontal burrows through the soil to move around and to feed and they will reuse these burrows to a certain extent. Endogeic earthworms are often pale colours, grey, pale pink, green or blue. Some can burrow very deeply in the soil.

Epigeic:

AN²

Earthworms live on the surface of the soil in leaf litter. These species tend not to make burrows but live in and feed on the leaf litter. Epigeic earthworms are also often bright red or reddy-brown, but they are not stripy.

African Night Crawlers (*Eudrilus-eugeniae*)



Source: Wikipedia

Epigeic earthworms live in areas containing high amounts of organic matter. They live at/near the soil surface and feed on leaf litter, decaying plant roots or dung. These earthworms do not form permanent burrows.

A worm's skin is photosensitive and therefore they need a dark environment. The **darkness** helps the worms to regulate their body temperature and moisture levels and provides them with a more stable and comfortable habitat.

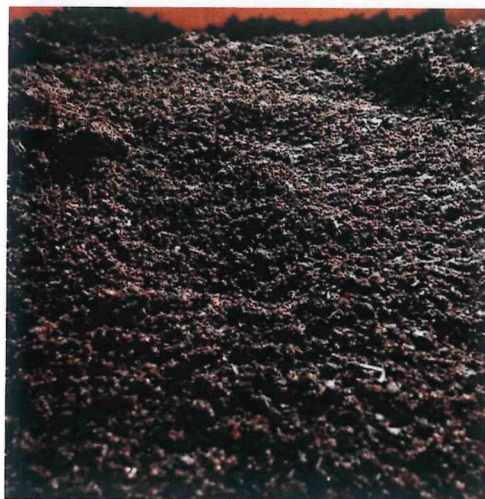
Facts:

- Voracious appetite – 1.5x body weight
- Able to feed on wide variety of organic waste
- Quick to reproduce
- Large – up to 8", plump up given more space
- Best in 21 – 29°C

These highly regarded specimens play a vital role in enhancing our soils and revitalizing the health of the plants, through their **Vermicast**.

What is Vermicast?

Vermicast (also called worm castings, worm humus, worm poop, worm manure, or worm faeces) is the end-product of the breakdown of organic matter by earthworms. It is usually found in vermicompost.



Source: www.rvarecycling.com

What is Vermicompost?

The worms consume most of the food and bedding & leave behind a mixture of their worm poop and undigested organic materials

Vermicompost may also contain worm eggs, thus introducing worms into the soil.

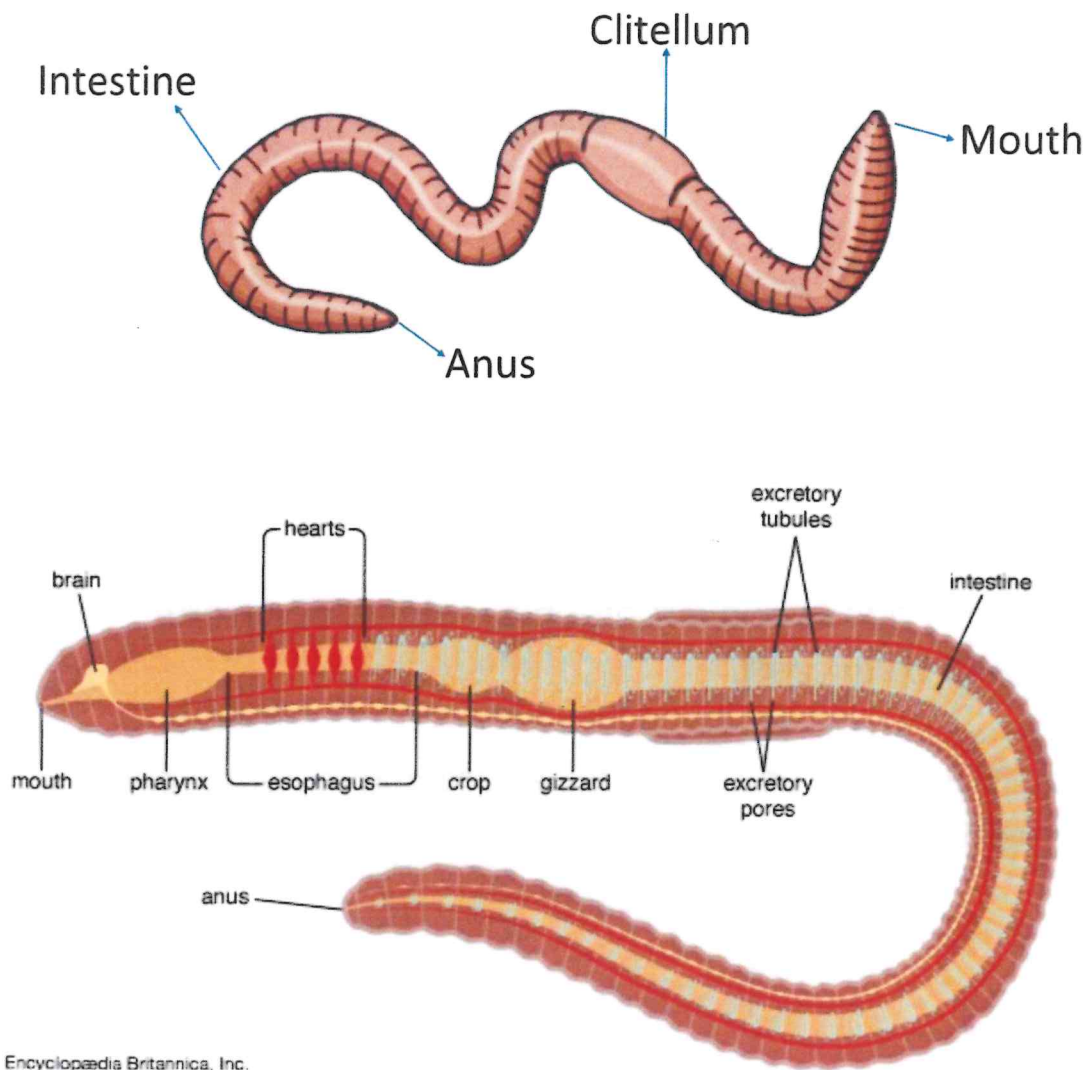
What is Vermicomposting?

Using earthworms to convert organic material into worm manure (aka vermicast).

It is a helpful way to use food waste (organic material) and create healthier soil.



Body Part:



Mouth

- The mouth of the African Night Crawler is located at the anterior end of its body.
- It serves as the entry point for ingesting organic matter and food particles from the environment.
- The mouth helps the worm to begin the process of digestion by taking in organic material, which is then processed further in the digestive system.

Clitellum

- The clitellum is a raised band located near the anterior end of the worm's body, typically closer to the head.
- It is involved in the reproductive process of the worm, particularly during sexual reproduction.
- The clitellum secretes a viscous, mucus-like substance that forms a protective cocoon for the eggs during egg deposition.
- In African Night Crawlers, the clitellum is an essential structure for the formation of cocoons where fertilization and embryonic development occur.
- **ANC are hermaphrodites.** It has both male and female reproductive organs but cannot fertilize itself. Hence you need to keep at least 2 worms for them to reproduce.

Intestine

- The intestine is a long, tubular organ that extends from the gizzard to the anus in the African Night Crawler.
- It is the primary site for nutrient absorption and digestion of food particles.
- Within the intestine, enzymes break down complex organic matter into simpler forms that can be absorbed by the worm's body.
- Nutrients from the digested food are absorbed through the intestinal wall and transported to various tissues for energy production, growth, and maintenance.

Anus

- The anus is the opening at the posterior end of the worm's body, responsible for expelling waste materials from the digestive system.
- Once digestion is complete and nutrients have been absorbed in the intestine, undigested and waste materials are passed through the anus as castings.

Why Vermicompost?

- Clean, sustainable and zero-waste
- Contains microbes
- Increases seed germination, vegetative growth of crops and yield
- Improves soil health

Vermicast vs Manure

	Vermicast	Manure
Odour	No	Yes
Need further processing	No	Yes
Ease of storage	Yes	No
Availability & accessibility	Yes	No

Promoting Soil Health

Vermicast contains a rich abundance of microbial life. The microorganisms/microbes produce a great amount of “gummy” substances like extracellular polysaccharides and mucilages that can cement soil particles together into **aggregates** or clusters.

The gummy substances make soil aggregates less likely to break apart when exposed to water, thereby increasing water infiltration and water holding capacity of the soil.

Vermicast can hold 2 - 3 times more water than their weight in soil. An aggregate is a naturally formed assemblage of sand, silt, clay, organic matter, plant root hairs, microorganisms, the glue-like secretions that microorganisms produce, and the hyphae of fungi (branching structures of certain fungus in the soil).

Vermicompost & Soil

Improves soil structure

In the realm of vermicompost, **soil structure** is profoundly enhanced through various interconnected mechanisms.

- **Enhanced Soil Aeration:**

Beneficial microbes within vermicompost decompose organic matter, fostering the creation of air-filled pores. This process facilitates gas exchange, promoting optimal conditions for root respiration, microbial activity, and

nutrient cycling. Improved soil aeration is pivotal for sustaining soil health and fostering robust plant growth.

- **Erosion Control and Soil Stability**

Vermicompost's beneficial microbes play a vital role in stabilizing soil aggregates and increasing soil organic matter content. By fortifying soil against erosion by water and wind, these microbes help maintain soil structure, fertility, and productivity over the long term.

- **Clay Breakdown and Root Penetration**

Certain microbes present in vermicompost, notably fungi and bacteria, produce enzymes that break down organic matter, effectively loosening clayey soil structures. This action improves soil tilth, facilitating enhanced soil aeration, drainage, and root penetration. Consequently, vermicompost promotes healthier plant growth by providing an environment conducive to root development.

- **Optimized Water Absorption**

Vermicompost's contribution to soil aggregation and pore space formation enhances water absorption. By allowing water to penetrate the soil more effectively and reducing surface runoff, vermicompost ensures adequate moisture levels for plant growth. This optimization mitigates the risks of waterlogging or drought stress, further bolstering plant health and vitality.

Enrich soil

Vermicompost, a boon to soil enrichment, works harmoniously with nature's intricate mechanisms to nurture thriving ecosystems.

- **Thriving Microbial Communities**

Within vermicompost, a rich diversity of beneficial microbes thrives, including bacteria, fungi, and other microorganisms. These microbial champions diligently decompose organic matter, cycle nutrients, and suppress harmful pathogens, fostering a balanced and fertile soil ecosystem conducive to robust plant growth.

- **Enhanced Nutrient Availability**

Vermicompost is a treasure trove of microbial activity, facilitating nutrient cycling processes essential for plant nourishment. Through the decomposition of organic matter, nitrogen fixation, and phosphorus solubilization, vermicompost enhances nutrient availability in forms readily accessible to plants. This heightened nutrient richness fortifies soil fertility, supporting vigorous plant growth and diminishing reliance on synthetic fertilizers.

- **pH Regulation for Optimal Nutrient Access**

The microbial magic within vermicompost extends to pH modulation, as certain microbes produce organic acids that gradually lower soil pH. This natural acidification process unlocks the availability of vital nutrients like phosphorus and micronutrients, promoting plant health and productivity by enhancing nutrient accessibility.

- **Nutrient Provision and Symbiotic Relations**

Vermicompost acts as a nurturing reservoir of nutrients, offering a bounty of organic and inorganic compounds transformed by microbial alchemy into forms readily assimilated by plants. Through symbiotic relationships with plant roots and nutrient-rich metabolic processes, vermicompost provides essential nourishment, fostering plant growth, enriching soil fertility, and championing sustainable agricultural practices.

Vermicompost & Plant Growth

Plant Growth Enhancement

- **Seeds germinate more quickly**

Vermicompost provides essential nutrients such as nitrogen, phosphorus, and potassium, along with beneficial microbial activity. These nutrients and microbes create optimal conditions for seed germination by promoting water uptake, nutrient availability, and hormone production, leading to faster emergence of seedlings.

- **Seedlings grow faster**

The nutrients and beneficial microorganisms present in vermicompost promote rapid growth and development of seedlings. Nutrients like nitrogen stimulate leaf and stem growth, while microbial activity enhances nutrient uptake efficiency. This synergy results in quicker establishment and robust growth of seedlings.

- **Root numbers and biomass increase**

Vermicompost enhances soil structure, providing a loose, well-aerated environment that encourages root growth and proliferation. Additionally, the nutrients and organic matter in vermicompost stimulate root development, resulting in increased root numbers and biomass. This extensive root system improves nutrient and water absorption, supporting healthier and more vigorous plants.

- **Earlier flowering of plants**

Vermicompost provides a balanced supply of nutrients essential for plant growth and development. Adequate nutrition promotes overall plant health, accelerating growth and leading to earlier flowering and fruiting stages. Additionally, the presence of beneficial microbes in vermicompost enhances nutrient uptake and hormone balance, further contributing to accelerated plant development.

Plant Health Improvement

- **Root stress tolerance improves**

Vermicompost contains compounds such as humic acids, enzymes, and plant growth regulators that enhance root health and function. These compounds strengthen root cell walls, improve water and nutrient absorption, and activate stress response mechanisms in plants. As a result, plants grown in vermicompost-enriched soil exhibit greater resilience to environmental stresses such as drought, disease, and transplant shock.

- **Reduces plant transplant shock**

The beneficial microbes and nutrients in vermicompost aid in reducing stress on plants during transplanting. Microbial activity promotes root growth and colonization, improving soil structure and nutrient availability. Nutrients in vermicompost also support overall plant vigor, enabling smoother transitions and reducing transplant shock.

- **Increases plant vitality and flavor profile**

Vermicompost enriches the soil with essential nutrients and promotes microbial activity, leading to healthier plants with improved vitality. Plants grown in vermicompost-enriched soil exhibit enhanced metabolic processes, including sugar and nutrient synthesis, resulting in improved flavor profiles in fruits and vegetables.

- **Increases leaf chlorophyll content**

Vermicompost supplies plants with essential nutrients such as nitrogen and magnesium, which are crucial for chlorophyll synthesis. Increased chlorophyll content in leaves improves photosynthetic efficiency, leading to healthier and more productive plants with greener and more vibrant foliage.

- **Suppress diseases in plants**

Certain beneficial microbes present in vermicompost act as natural antagonists to plant pathogens. These microbes outcompete harmful pathogens for resources, produce antibiotics or other compounds toxic to pathogens, and stimulate plant defenses. By suppressing diseases, vermicompost helps maintain plant health and productivity without the need for chemical pesticides,

contributing to sustainable agricultural practices.

Hot Composting



Hot composting is a method of composting organic materials that relies on microbial activity to generate high temperatures within the compost pile. This accelerated decomposition process results in the rapid breakdown of organic matter into nutrient-rich compost.

In hot composting, a mix of green (nitrogen-rich) and brown (carbon-rich) materials is layered or mixed together in a pile or bin. The green materials provide nitrogen, which serves as a protein source for microorganisms, while the brown materials provide carbon, acting as an energy source.

As microorganisms break down the organic matter, they generate heat as a byproduct of their metabolic processes. The

internal temperature of the compost pile can rise significantly, reaching temperatures between 130°F to 160°F (55°C to 71°C) or even higher.

Maintaining these high temperatures is essential for hot composting because they accelerate the decomposition process and help kill weed seeds, pathogens, and undesirable microorganisms. To achieve and sustain high temperatures, hot compost piles must be properly aerated, with adequate moisture levels and a balanced carbon-to-nitrogen ratio.

Hot composting typically results in compost ready for use in a shorter time frame compared to other composting methods, usually within a few weeks to a few months, depending on factors such as pile size, materials used, and environmental conditions. The resulting compost is rich in nutrients and beneficial microorganisms, making it an excellent soil amendment for improving soil fertility and plant growth.

Materials required

- **Green (Nitrogen-rich) Materials:**
Such as kitchen scraps, grass clippings, and fresh garden waste. These provide nitrogen, which aids in the decomposition process.
- **Brown (Carbon-rich) Materials:**
Like dry leaves, straw, and shredded paper. These provide carbon, serving as an energy source for microorganisms.
- **Moisture:**
Composting organisms require moisture to thrive, but the pile shouldn't be too wet. Aim for a consistency similar to a damp

sponge.

- **Aeration:**

Turning the compost pile regularly helps maintain aerobic conditions, ensuring proper airflow and preventing anaerobic decomposition (which can lead to unpleasant odors).

- **Composting Bin:**

Enclosed container or designated composting area to contain the materials and facilitate decomposition.

- **Thermometer:**

Optional but helpful for monitoring the internal temperature of the compost pile to ensure it reaches the optimal range for hot composting (130°F to 160°F or 55°C to 70°C).

Considerations

- **Size of Materials:**

Chop or shred larger organic waste items to speed up decomposition and increase surface area.

- **Optimal Volume:**

A minimum pile size (around 3 feet in each dimension) is necessary to generate and maintain high temperatures. Larger piles retain heat better.

- **Temperature Monitoring:**

Hot composting relies on high internal temperatures (ideally between 130°F to 160°F or 55°C to 71°C) to speed up decomposition and kill pathogens and weed seeds.

- **Turning Frequency:**

Turn the compost pile regularly (every few days to a week) to aerate and mix the materials, promoting even decomposition and preventing anaerobic conditions.

- **Odor Control:**

Properly manage the compost pile to minimize odors by ensuring adequate aeration and avoiding adding materials that may cause foul smells.

- **Carbon-to-Nitrogen Ratio (C:N):**

Maintaining a balanced C:N ratio (around 25-30 parts carbon to 1 part nitrogen) ensures efficient decomposition. Too much nitrogen can lead to odors, while too much carbon can slow down decomposition.

- **Patience:**

Hot composting requires time and effort. Be patient and consistent with maintenance tasks to achieve high-quality compost efficiently.

By ensuring these components and conditions are met, hot composting can efficiently break down organic materials into nutrient-rich compost in a relatively short period, typically within a few weeks to a few months.

Advantages of Vermicomposting (over Composting)

- Requires less space
- Consumes less time & produces compost faster
- Requires less labour, less maintenance

Setting up a worm bin



Type of bin

Selecting the right type of bin is essential for successful vermicomposting.

Common options include plastic containers, wooden bins, or specialized vermicompost bins available commercially.

Consider factors such as durability, **opacity**, and ease of access for harvesting compost and managing the worms.

Ensure the bin has a tight-fitting lid to maintain proper moisture levels and prevent pests from entering. Worms typically are sensitive to light so you might consider an opaque bin to ensure

there is a high degree of darkness within the bin.

Size of bin

The size of the bin depends on the amount of organic waste you generate and the space available for composting.

A larger bin accommodates more organic materials and worms, allowing for higher composting capacity. However, larger bins may require more space and maintenance.

Consider the volume of waste produced regularly and the number of worms needed to process it efficiently.

Aim for a size that meets your composting needs while fitting within your available space.

Choice of worms:

The type of worms used in vermicomposting is crucial for its success.

The most common species used are, given our local climate, **African Night Crawlers and Malaysian Blue.**

These worms are voracious eaters, adapt well to composting conditions, and reproduce quickly.

When choosing worms, consider factors such as availability, suitability for your climate, and compatibility with the materials you'll be composting.

Purchase worms from reputable sources to ensure they are healthy and free from diseases or pests.

By considering these factors when setting up a vermicompost bin, you can create an optimal environment for efficient decomposition of organic waste and the production of nutrient-rich compost by your worm colony.

African Night Crawlers

Larger in size

African Night Crawlers are typically larger in size. They can grow up to 8" long. This characteristic allows them to process larger quantities of organic waste efficiently. Their larger size also contributes to their ability to burrow through the compost pile, facilitating aeration and decomposition.

Voracious eater

African Night Crawlers have a voracious appetite, meaning they consume organic matter at a rapid rate. They can eat up to 1.5 times their body weight. This trait makes them highly efficient composters, as they quickly break down a wide variety of organic materials, including kitchen scraps, yard waste, and paper. Their efficient digestion process helps convert organic waste into nutrient-rich castings, which are valuable for soil health and plant growth.

Rapid reproduction

These African night crawlers grow fast and will reach their full size by around 8-10 weeks old. They also reproduce fast and can be bred at about 5 weeks old. The eggs take 1 to 2 weeks to hatch. They produce about 10.5 baby worms/week. There are 2-3 worms/eggs.

High adaptability

African Night Crawlers exhibit a remarkable ability to adapt to various environmental conditions. They can tolerate a wide range of temperatures (21 – 30 degC) , moisture levels, and pH levels, making them well-suited for vermicomposting in diverse climates and settings. This adaptability ensures that African Night Crawlers can thrive and function effectively in different composting environments, contributing to the overall success of the vermicomposting process.

Malaysia Blue

- Origin: Malaysia
- Small and thin, not more than 3“ long
- Difficult to harvest their castings due to the worms' small size
- Escape artists with a propensity for escaping in the hundreds if conditions are not favourable

Malaysian Blue vs ANC



Worm & Bin Care

What to feed:

Browns

The two main things that microorganisms need are not actually carbon and nitrogen. They need protein to grow & reproduce (protein is nitrogen-rich) and sugar for energy (carbon materials tend to be sugar/carb-rich).

Characteristics of brown items: rich in carbon, dry, fibrous

Types of Browns:

- Shredded cardboard - doubles as food and bedding.
Worms can survive entirely on cardboard as it is safe for them.
- Paper egg carton
- Newspaper
- Shredded paper – avoid glossy paper, coloured ink & used tissue
- Dried leaves

Greens

Characteristics of green items: rich in nitrogen and tend to be wet (give moisture)

Types of Greens

- Vegetable waste (carrots, lettuce, beans, peas, limited amounts of potatoes, leaf vegetables)
- Non-citrus fruit waste (apples, grapes, bananas, plums, peaches, pumpkin)
- Coffee grounds (filters too) – help to ward off ants.
Decaffeinated coffee is less nutritional for worms as some

sugars and proteins are also removed with the caffeine.

Others

Eggshells

- **Dried** eggshells are not considered a brown or a green in composting because they are primarily mineral, not carbon or nitrogen.
- Powdered eggshells help with digestion (grit).
- Acts as a pH buffer when bedding becomes too acidic .
- Eggshells are believed to be helpful for worm reproduction (worms have been observed to hide and lay their eggs in whole/halved eggshells).

WHAT NOT TO FEED:

Citrus fruits (orange, lemon peels) – acidic foods can cause the bedding to become over-acidic and kill the worms

Meat, dairy products, oily foods, animal waste, human waste – stink, toxic and attract vermin

Onion, garlic, chilli - the compounds found in each of them irritates the worms

Seeds – will not be broken down even after they have germinated as worms avoid living plants and prefer decomposing material

Bin Management

- The worms can go straight into a new bin with just browns.
- There is a maximum density of worms in a box – when they reach max density the worms become smaller and maybe even start losing their clitellum.
- A low density of worms is around 20 worms.

- What they do: 2 months pre-compost, 2 months of worms in the bin, 1 month for the eggs to hatch.
- After the last month move the worms to a new bin.

Worm Eggs & Juveniles



Bin Maintenance

Moisture and Oxygen

- Counteract each other
- High moisture block oxygen flow
- High air-flow increases evaporation
- Balanced - like a wrung out sponge
- Spray water if dry; too wet, add Browns

Temperature

- Ensure airflow
- Always ensure that there is a cool area
- Do not overfeed
- Do practice rotating the feeding area

Bedding

- carbon source to balance nitrogen
- Adds bulk to the bin to prevent matting and increase airflow.
- Absorbs excess moisture.

Light

- very sensitive to UV light
- Place bin in shady area

Feeding Frequency & Quantity

- Overfeeding the worms can kill them & make the bin very hot
- Feed them a small amount of greens (chopped/blended) once a week – 1/4 of the surface area

Worm's Diet

ALWAYS

Vegetables
Fruits
Shredded newspapers
Soggy cardboards

SOMETIMES

Coffee grounds
Used tea bags
Eggshells *powdered*
Legumes *etc*

NEVER

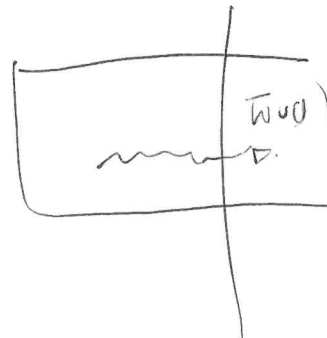
Citrus
Peppers
Onions & Garlic
Animal products
Oily food

cut their delicate stem.

Harvesting

By Migration

- Add food to one end of the bin.
- Wait a day.
- Harvest from the other end

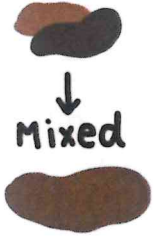


By Light

- shine light at a pile of castings on one side of bin
- wait for a day
- they will move away from the light
- Harvest pile under light

Using Vermicompost & Compost Tea

24h.

Soil Amendment	Top dressing	Seed Starter	liquid fertilizer (Compost tea)
 Mixed Mix potting/garden soil with vermicompost before planting	 Sprinkle above soil. No measuring or mixing required.	 Incorporate vermicompost into seed starting mix.	 1 cup vermicompost 0.5 tbs molasses 2 litres water Steep overnight and water plants once every 10 days.

won't attract ants
-: microbes
bigger from by microbes

Compost Tea *Bacteria culture.*

1. Put 2 litres of rainwater in a pail
2. Add 1 cups of vermicompost
3. Add 0.5 tbs molasses (to grow healthy microbes)
4. Bubble with airstone for 48h until dark brown
5. Dilute till light brown (teh O po)
6. Use (within 24h)

Fermentation



Worm Tower



In a worm tower, it is less about making the perfect living space and more about providing a feeding zone.

Cross-section View



Advantages

- Low maintenance
- Worms can leave the tower if food goes wrong (e.g. too wet/dry/hot/acidic) and return when the food has composted enough
- No harvesting required - worms deposit worm castings around the garden, aerating the soil in the process

Troubleshooting

There are fruit flies. What should I do?

Bury or cover the food scraps properly.

Alternatively, you may apply a small amount of BTI (*Bacillus thuringiensis* subsp. *israelensis*) onto compost.

There is a strong, unpleasant odour.

Reduce the amount you are feeding as the wastes are not being consumed quickly and decomposition has set in.

Worms are escaping!

This is a sign that the bedding is acidic or too wet and is not favourable for the worms to thrive in.

To resolve the issue, avoid feeding the worms acidic food scraps and ensure that the bedding is moist but not wet by adding paper scraps.

Testing

Worms escaped and my household pet ate them!

The African Night Crawler is a non-poisonous species. The worms will not cause harm to your pets if they consume any.

How quickly will I see results?

A month. Vermicomposting is one of the fastest ways of converting bulky waste products into compact nutrient-dense black gold.

Give us Feedback!



- Egg 12-16 day
- Hatch to maturity 5-7 wks
- # cocoons
- Lifespan 3 yrs

